

Smart Energy Monitoring System with Real-Time Power Measurement

Senior Project Final Report

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Date: May 2026



Abstract

This project focused on designing, implementing, and testing a Smart Energy Monitoring System capable of measuring real-time electrical parameters such as voltage, current, power, energy, and frequency. The system was built using an ESP32 microcontroller, a PZEM-004T power monitoring module, and a current transformer (CT) sensor. The project emphasized embedded system integration, AC power measurement, serial communication, and electrical system safety. The main goal was to develop a low-cost, functional monitoring system that demonstrates practical electrical engineering concepts while applying knowledge gained throughout undergraduate coursework. The final prototype successfully measured and displayed real-time electrical data from an AC-powered load. During development, several challenges arose involving AC wiring, sensor integration, and communication protocols. These issues required troubleshooting, debugging, and iterative design improvements. This project provided hands-on experience in electrical measurements, embedded system design, hardware integration, signal processing, and engineering problem-solving. It also showed how concepts learned in the Electrical and Computer Engineering curriculum can be applied to solve practical engineering problems.

1. Introduction

Electrical energy monitoring systems are becoming increasingly vital in modern residential, commercial, and industrial settings. Monitoring electrical parameters such as voltage, current, power consumption, and frequency helps users better understand energy use, boost efficiency, and detect abnormal electrical conditions.

The goal of this project was to design and implement a Smart Energy Monitoring System capable of measuring and displaying electrical parameters in real time. The system was built using an ESP32 microcontroller and a PZEM-004T energy monitoring module. The ESP32 handled data processing and displayed results through serial communication.

A key objective of the project was not only to create a working system but also to apply engineering concepts learned throughout undergraduate studies. This project integrated principles from circuits, electronics, embedded systems, communication systems, programming, and electrical measurements into a single system.

2. Project Objectives

The primary objectives of the project were:

- Design a system capable of measuring voltage, current, power, energy, and frequency.
 - Interface an ESP32 microcontroller with a PZEM-004T measurement module.
 - Safely integrate AC power into the system.
 - Display real-time electrical measurements.
 - Demonstrate embedded systems communication using UART.
 - Apply engineering principles learned during undergraduate studies.
 - Develop troubleshooting and system integration skills.
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3. Engineering Knowledge Applied Throughout the Project

One of the most important aspects of this project was the application of concepts learned throughout my collegiate years in the Electrical and Computer Engineering program. This project required me to combine both theoretical and practical knowledge from multiple courses.

3.1 Circuit Analysis and Electrical Principles

Fundamental electrical engineering concepts learned in circuit analysis courses played a major role throughout the project. Concepts such as Ohm's Law, Kirchhoff's Laws, AC circuit behavior, and power calculations were essential for understanding how the monitoring system operated.

The relationship between voltage, current, and power was continuously used during testing and validation. For example, measured power values were compared to the theoretical equation:

$$P = VI$$

This relationship was used to verify that the system measurements were physically reasonable.

Knowledge of AC systems was especially important because the project involved live AC power. Understanding RMS voltage, frequency, sinusoidal waveforms, and safe wiring practices helped ensure proper system operation.

3.2 Embedded Systems and Microcontrollers

Courses involving embedded systems and programming were directly applied during implementation. The ESP32 microcontroller served as the central processing unit of the system. Programming knowledge learned through C++ and embedded systems courses was used to:

- Configure UART communication.
- Read data from the PZEM module.
- Process electrical measurements.
- Output formatted data through the serial monitor.
- Troubleshoot communication errors.

The project reinforced the importance of debugging and software-hardware integration.

3.3 Communication Systems

The project also applied communication concepts through UART serial communication. The ESP32 communicated with the PZEM module using TX and RX lines at a baud rate of 9600. Understanding asynchronous communication protocols, baud rates, and data transmission formats was necessary for establishing stable communication between devices.

This portion of the project demonstrated how communication protocols are used in real embedded systems.

3.4 Electronics and Sensors

Knowledge gained from electronics courses was applied when working with sensors and hardware integration. The current transformer used in the system operates using electromagnetic induction principles. Understanding how current generates a magnetic field helped explain why the CT clamp must surround only one conductor.

The project also reinforced the importance of signal measurement and electrical isolation when interfacing sensors with high-voltage systems.

3.5 Engineering Problem Solving

Throughout the project, engineering problem-solving skills were constantly required. Multiple hardware and software issues occurred during implementation, including:

- Incorrect wiring configurations.
- Invalid sensor readings.
- UART communication errors.
- Current transformer placement problems.
- Hardware damage caused during testing.

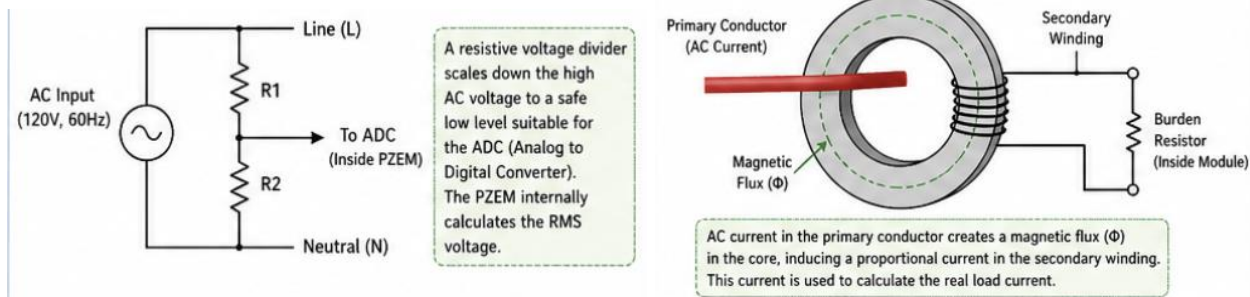
Solving these issues required systematic troubleshooting methods learned throughout laboratory courses and engineering design experiences.

4. System Design

The system consisted of three major components:

1. PZEM-004T measurement module.
2. Current transformer sensor.
3. ESP32 microcontroller.

The PZEM module was responsible for measuring electrical parameters from the AC circuit. The current transformer measured current non-invasively by detecting the magnetic field around the hot conductor. The ESP32 processed measurement data and displayed the results.

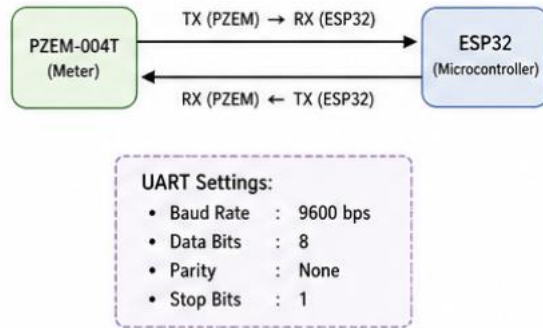


Voltage Measurement Principle

CT Sensor Working Principle

The AC input from the wall supply was connected to the PZEM module. The load, which consisted primarily of a lamp during testing, was connected to the output side of the module. The CT clamp was placed around the hot wire only.

The ESP32 communicated with the PZEM module through UART serial communication. The TX and RX pins were connected between the two devices to allow real-time data exchange.



Data Communication Diagram

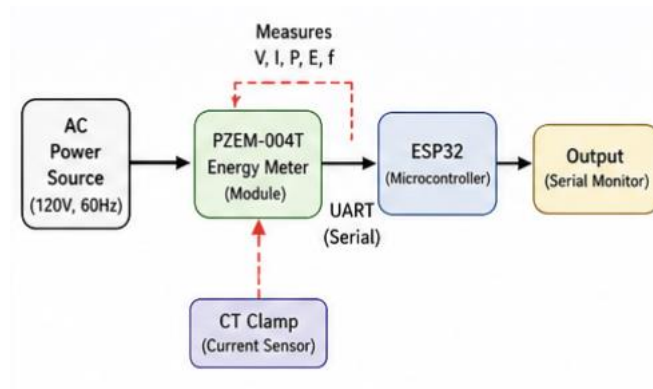
5. Hardware Components

The major hardware components used in this project included:

- ESP32 development board.
- PZEM-004T-100A V4.0 power monitoring module.
- Current transformer clamp.
- Jumper wires.
- Extension cord wiring.
- Lamp used as test load.
- USB cable for programming and power.

The ESP32 was selected because of its processing capabilities, simplicity, and compatibility with Arduino IDE development tools.

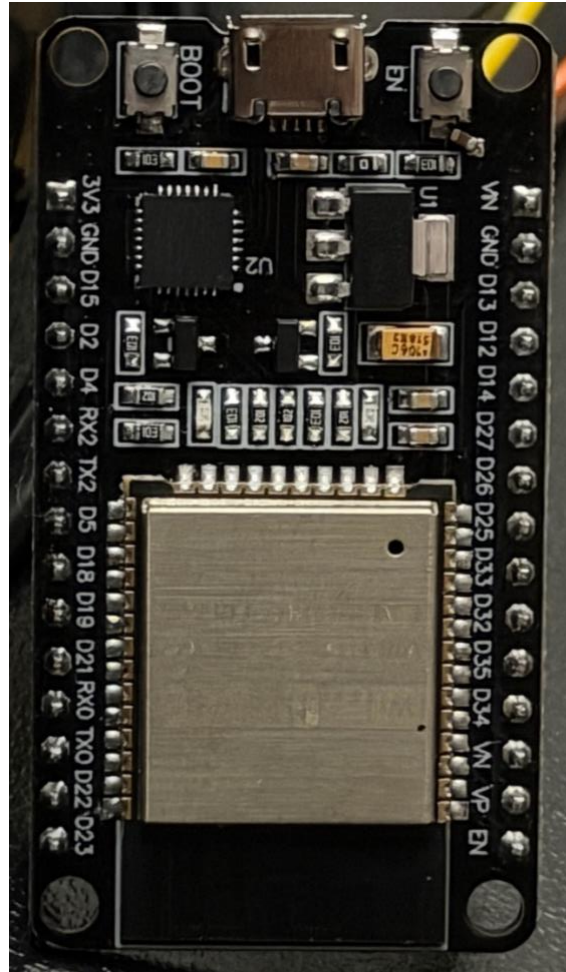
The PZEM-004T module was selected because it provided integrated electrical measurement capabilities for voltage, current, power, frequency, and energy.



Block diagram of Smart Energy Monitoring System



The PZEM-004T



The ESP32



The CT Clamp

6. Software Design and Implementation

The software portion of the project was developed using Arduino IDE and programmed in C++. The program first initialized serial communication for both the serial monitor and the PZEM communication interface. UART communication was configured at 9600 baud to enable communication with the PZEM module. The ESP32 repeatedly requested electrical measurements from the PZEM module and displayed the results in real time. These measurements included voltage, current, power, energy, and frequency. The program also included formatted serial monitor output to improve readability during demonstrations.

6.1 Arduino IDE C++ code:

```
#include <PZEM004Tv30.h>

// Use ESP32 UART1
HardwareSerial mySerial(1); // UART1

// PZEM object
PZEM004Tv30 pzem(mySerial, 16, 17);

void setup() {
    Serial.begin(115200);

    // Initialize UART1
    mySerial.begin(9600, SERIAL_8N1, 16, 17);

    Serial.println("PZEM004Tv30 Monitoring Started");
}

void loop() {

    float voltage = pzem.voltage();
    float current = pzem.current();
    float power   = pzem.power();
    float energy  = pzem.energy();
    float freq    = pzem.frequency();

    // Remove invalid readings
```



```
if (isnan(voltage) || voltage < 50) {  
    voltage = 0;  
    current = 0;  
    power = 0;  
    energy = 0;  
    freq = 0;  
}  
  
Serial.println("===== POWER MONITOR =====");  
  
Serial.print("Voltage: ");  
Serial.print(voltage);  
Serial.println(" V");  
  
Serial.print("Current: ");  
Serial.print(current);  
Serial.println(" A");  
  
Serial.print("Power: ");  
Serial.print(power);  
Serial.println(" W");  
  
Serial.print("Energy: ");  
Serial.print(energy);  
Serial.println(" Wh");  
  
Serial.print("Freq: ");  
Serial.print(freq);  
Serial.println(" Hz");  
  
Serial.println("=====\n");  
  
delay(2000);  
}
```

7. Project Progress and Development Process

The project progressed through several stages, including research, component selection, hardware integration, software development, testing, debugging, and validation.

7.1 Initial Research and Planning

The project began with researching methods for electrical energy monitoring using embedded systems. Multiple approaches and measurement modules were evaluated before selecting the PZEM-004T module.

During this stage, system requirements and safety considerations were also analyzed.

7.2 Component Integration

After receiving the components, the ESP32 and PZEM module were connected through UART communication. Initial software development focused on establishing stable communication between the devices.

Several issues were encountered during this phase involving incorrect baud rate configurations and library compatibility.

7.3 Hardware Testing

Once communication was established, the system was connected to a low-power AC load for testing.

Initially, the system only produced voltage and frequency measurements while current and power remained at zero. Troubleshooting revealed that the current transformer placement was incorrect.

The CT clamp was eventually placed correctly around the hot wire only, which allowed the system to successfully measure current and power.

7.4 Hardware Failure and Recovery

During testing, one of the PZEM modules was accidentally damaged due to wiring mistakes while working with AC power.

Although this setback delayed progress temporarily, the issue reinforced the importance of electrical safety and proper wiring validation. A replacement module was ordered, and testing resumed after correcting the wiring configuration.

This experience significantly improved my understanding of real-world electrical system integration and safe testing procedures.

7.5 Final Validation

After correcting the wiring and CT placement, the system successfully produced stable and realistic electrical measurements.

Typical measurements included:

- Voltage ≈ 122 V.
- Current ≈ 0.20 A.
- Power ≈ 24.6 W.
- Frequency ≈ 60 Hz.

These results were validated using theoretical calculations.

8. Challenges Encountered

Several technical and implementation challenges were encountered throughout the development of the Smart Energy Monitoring System. These challenges occurred during both the hardware integration and software development stages of the project and required extensive troubleshooting, testing, and redesign efforts. Although these issues temporarily slowed progress, they significantly improved my understanding of embedded systems, electrical measurements, AC power systems, and engineering problem-solving methods.

One of the first major challenges involved correctly integrating the AC wiring with the PZEM-004T monitoring module. Since the system operated using live AC voltage, proper placement of the hot and neutral conductors was critical for both functionality and safety. During initial testing, incorrect terminal connections resulted in invalid voltage readings, including repeated “NaN” outputs from the module. These errors occurred because the PZEM module relies on a proper line-to-neutral reference to perform accurate voltage measurements. Through troubleshooting and reviewing the module documentation, the wiring configuration was corrected, and stable voltage measurements were eventually achieved.

Another significant challenge involved the current transformer (CT) clamp configuration. Initially, the CT clamp was incorrectly placed around both the hot and neutral conductors simultaneously. Because AC flows in opposite directions through the hot and neutral wires, the magnetic fields generated by each conductor cancel one another inside the transformer core. As a

result, the CT sensor measured nearly zero current even when the electrical load was active. This issue demonstrated the importance of understanding electromagnetic induction and transformer operation. After analyzing the problem, the CT clamp was repositioned around only the hot conductor, which immediately allowed the system to correctly detect current flow and calculate power consumption.

Communication issues between the ESP32 and the PZEM module also created challenges during development. The system used UART serial communication, which required correct baud rate configuration and proper TX/RX pin connections. Early in development, incorrect baud rate settings and unstable jumper wire connections caused corrupted data, failed communication, and inconsistent readings. Additional debugging revealed that the PZEM module communicated at 9600 baud while the serial monitor operated at 115200 baud. Once the communication settings were configured correctly and wiring stability improved, reliable serial communication was established.

Hardware reliability also became a challenge during testing. While working with AC power connections, one of the monitoring modules was accidentally damaged due to a wiring mistake. This required the replacement of the module and temporarily delayed testing progress. Although this setback was frustrating, it reinforced the importance of careful electrical verification, proper isolation techniques, and safe engineering practices when working with live power systems. This experience also improved my confidence in troubleshooting hardware failures and rebuilding the system correctly.

Additional challenges appeared during the software validation stage of the project. During testing, the PZEM module occasionally produced invalid readings whenever the AC system was disconnected or powered off. In several cases, the voltage reading returned “NaN” while the current and power values continued displaying stale values from previous measurements. This behavior occurred because the sensing circuitry no longer had a valid AC reference, causing unstable or floating sensor outputs. As a result, the displayed values no longer accurately represented the actual state of the system.

To improve measurement reliability, additional software validation logic was implemented within the ESP32 program. An if-statement was added to detect invalid voltage measurements before displaying data to the user. If the voltage reading returned “NaN” or fell below a minimum threshold value of 50 V, all electrical measurements were automatically reset to zero. The implemented logic was:

```
if (isnan(voltage) || voltage < 50) {  
    voltage = 0;  
    current = 0;  
    power = 0;  
    energy = 0;  
    freq = 0;  
}
```

This software improvement demonstrated the importance of signal validation, noise filtering, and fault detection in embedded systems engineering. In practical applications, sensors and communication systems can produce unstable or misleading values when operating outside

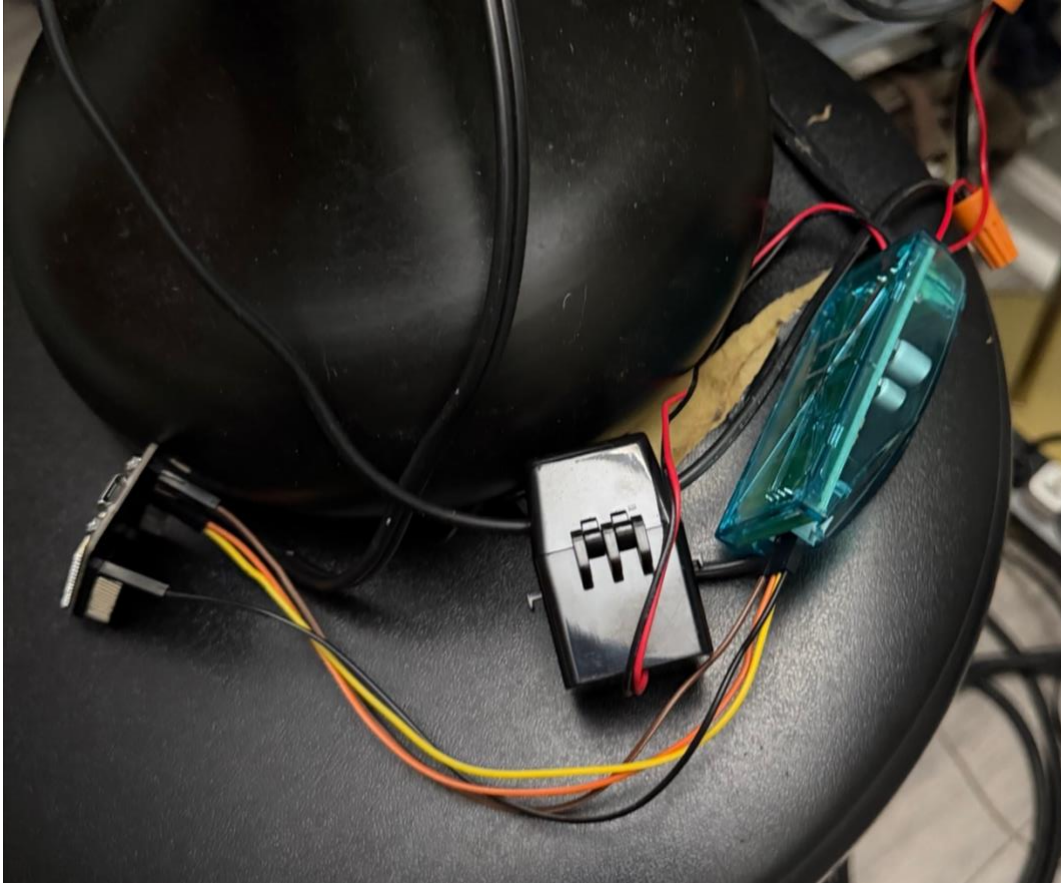
normal conditions. Implementing threshold-based validation improved the reliability and professionalism of the system by preventing invalid measurements from being displayed during disconnected or powered-off states.

The threshold value of 50 V was selected because the system normally operates near standard residential AC voltage levels of approximately 120 V. Any measured voltage significantly below this range was considered an invalid operating condition. This design choice provided a simple but effective method for filtering out false or unstable sensor readings.

Overall, each challenge encountered during the project contributed to a deeper understanding of electrical engineering concepts, embedded systems integration, debugging techniques, and real-world engineering design. The troubleshooting process strengthened my analytical thinking, patience, and ability to systematically solve technical problems under practical conditions.

9. Results

After correcting the wiring configuration and properly positioning the current transformer around the hot conductor, the system successfully produced stable and realistic electrical measurements. The measurements remained consistent throughout repeated testing, indicating reliable communication between the ESP32 and the PZEM-004T module as well as proper operation of the sensing hardware.



Correctly wired smart energy monitoring system

Typical measured values during testing included:

- Voltage ≈ 122 V
- Current = 0.20 A
- Power ≈ 24.6 W
- Frequency ≈ 60 Hz

These measured values were validated using theoretical electrical calculations and by comparing them to standard residential electrical characteristics in the United States.

In the United States, standard residential electrical outlets typically provide approximately 120 V AC at a frequency of 60 Hz. Due to utility tolerances and load variations, measured voltages commonly range between approximately 110 V and 125 V under normal operating conditions. The measured voltage of approximately 122 V therefore fell well within the expected range for standard household AC power systems, confirming that the voltage measurement portion of the monitoring system was functioning properly.

The test load used during validation was a 25 W incandescent light bulb. Because incandescent bulbs behave primarily as resistive loads, they provide a predictable electrical response and are

useful for validating measurement systems. For resistive loads, the power factor is approximately equal to 1, allowing the electrical power relationship to be simplified to:

$$P = VI$$

Where:

- P = power in watts
- V = voltage in volts
- I = current in amperes

Using the measured voltage of approximately 122 V and the rated bulb power of 25 W, the theoretical current draw can be calculated by rearranging the equation:

$$I = \frac{P}{V}$$

Substituting the known values:

$$I = \frac{24.6W}{122V}$$
$$I \approx 0.202A$$

The measured current from the monitoring system was approximately:

$$I_{measured} = 0.20A$$

The measured current closely matched the theoretical value, with only a very small difference caused by normal measurement tolerances, electrical fluctuations, and sensor accuracy limitations.

The percent error between the theoretical and measured current values was calculated using:

$$\% \text{ Error} = \left| \frac{\text{Measured} - \text{Theoretical}}{\text{Theoretical}} \right| \times 100$$

Substituting the values:

$$\% \text{ Error} = \left| \frac{0.20 - 0.202}{0.202} \right| \times 100$$
$$\% \text{ Error} \approx 0.99\%$$

This very small percent error demonstrated that the measured current closely matched the theoretical current value predicted by electrical power equations.

The measured power value was also validated using the measured voltage and current:

$$P = VI$$

Substituting the measured values:

$$\begin{aligned} P &= (122V)(0.20A) \\ P &= 24.4 \text{ W} \end{aligned}$$

The measured system power value was approximately:

$$P_{\text{measured}} \approx 24.6W$$

The percent difference between the calculated and measured power values was determined using:

$$\begin{aligned} \% \text{ Difference} &= \left(\frac{|24.6 - 24.4|}{\frac{24.6 + 24.4}{2}} \right) \times 100 \\ &= \left(\frac{0.2}{24.5} \right) \times 100 \\ &\approx 0.82\% \end{aligned}$$

This extremely small percent difference demonstrated strong agreement between the theoretical calculations and the actual measured power values.

The measured frequency also remained consistently near 60 Hz, which matches the standard AC power frequency used throughout the United States electrical grid. Stable frequency readings further confirmed that the system was properly detecting and processing the AC waveform.

Overall, the close agreement between theoretical calculations, expected household electrical standards, and measured system outputs demonstrated that the Smart Energy Monitoring System was functioning correctly and producing valid electrical measurements.


```

06:53:10.373 -> Voltage: 122.10 V
06:53:10.373 -> Current: 0.20 A
06:53:10.373 -> Power: 24.60 W
06:53:10.373 -> Energy: 0.01 Wh
06:53:10.373 -> Freq: 60.00 Hz
06:53:10.373 -> =====
06:53:10.373 ->
06:53:12.376 -> ===== POWER MONITOR =====
06:53:12.376 -> Voltage: 122.10 V
06:53:12.376 -> Current: 0.20 A
06:53:12.376 -> Power: 24.60 W
06:53:12.376 -> Energy: 0.01 Wh
06:53:12.376 -> Freq: 60.00 Hz
06:53:12.376 -> =====
06:53:12.376 ->
06:53:14.373 -> ===== POWER MONITOR =====
06:53:14.373 -> Voltage: 122.20 V
06:53:14.373 -> Current: 0.20 A
06:53:14.406 -> Power: 24.60 W
06:53:14.406 -> Energy: 0.01 Wh
06:53:14.406 -> Freq: 59.90 Hz
06:53:14.406 -> =====
06:53:14.406 ->
06:53:16.405 -> ===== POWER MONITOR =====
06:53:16.405 -> Voltage: 122.20 V
06:53:16.405 -> Current: 0.20 A
06:53:16.405 -> Power: 24.70 W
06:53:16.405 -> Energy: 0.01 Wh
06:53:16.405 -> Freq: 60.00 Hz
06:53:16.405 -> =====
06:53:16.405 ->
06:53:18.396 -> ===== POWER MONITOR =====
06:53:18.396 -> Voltage: 122.30 V
06:53:18.396 -> Current: 0.20 A
06:53:18.429 -> Power: 24.60 W
06:53:18.429 -> Energy: 0.01 Wh
06:53:18.429 -> Freq: 60.00 Hz
06:53:18.429 -> =====

```

The serial monitor display while the lamp is on.

```

06:53:24.445 -> Voltage: 122.30 V
06:53:24.445 -> Current: 0.20 A
06:53:24.445 -> Power: 24.60 W
06:53:24.445 -> Energy: 0.01 Wh
06:53:24.445 -> Freq: 59.90 Hz
06:53:24.445 -> =====
06:53:24.445 ->
06:53:26.547 -> ===== POWER MONITOR =====
06:53:26.547 -> Voltage: 0.00 V
06:53:26.547 -> Current: 0.00 A
06:53:26.547 -> Power: 0.00 W
06:53:26.547 -> Energy: 0.00 Wh
06:53:26.580 -> Freq: 0.00 Hz
06:53:26.580 -> =====
06:53:26.580 ->
06:53:28.670 -> ===== POWER MONITOR =====
06:53:28.670 -> Voltage: 0.00 V
06:53:28.670 -> Current: 0.00 A
06:53:28.670 -> Power: 0.00 W
06:53:28.670 -> Energy: 0.00 Wh
06:53:28.670 -> Freq: 0.00 Hz
06:53:28.670 -> =====
06:53:28.670 ->
06:53:30.766 -> ===== POWER MONITOR =====
06:53:30.766 -> Voltage: 0.00 V
06:53:30.766 -> Current: 0.00 A
06:53:30.766 -> Power: 0.00 W
06:53:30.766 -> Energy: 0.00 Wh
06:53:30.799 -> Freq: 0.00 Hz
06:53:30.799 -> =====
06:53:30.799 ->
06:53:32.897 -> ===== POWER MONITOR =====
06:53:32.897 -> Voltage: 0.00 V
06:53:32.897 -> Current: 0.00 A
06:53:32.897 -> Power: 0.00 W
06:53:32.897 -> Energy: 0.00 Wh
06:53:32.897 -> Freq: 0.00 Hz
06:53:32.897 -> =====

```

The serial monitor display when the lamp is switched off.

10. Overall Experience and Reflection

This project provided valuable hands-on engineering experience and reinforced key concepts learned throughout my undergraduate studies. One of the most important lessons was the gap between theoretical knowledge and real-world implementation, as integrating hardware and software into a functional system introduced additional complexity beyond what is typically seen in coursework.

The project strengthened my understanding of embedded systems, electrical measurements, AC circuit behavior, communication protocols, troubleshooting methods, and engineering safety. It also improved my confidence in designing and debugging complete engineering systems.

Although several setbacks occurred during development, overcoming these challenges significantly enhanced my problem-solving abilities. Overall, the project effectively demonstrated how knowledge gained throughout my academic studies can be applied to develop a practical and functional engineering solution.

11. Future Improvements

Potential future improvements for the Smart Energy Monitoring System include adding an LCD or OLED display for standalone real-time visualization, implementing wireless monitoring capabilities using Wi-Fi or Bluetooth, incorporating long-term data logging functionality, developing a mobile or web application interface, expanding the system to support multi-load monitoring, and designing a more professional enclosure and packaging system. These improvements would significantly increase the overall functionality, usability, and marketability of the product by making the system more user-friendly, portable, and visually appealing for both residential and commercial applications.

12. Conclusion

The Smart Energy Monitoring System project demonstrated the design and implementation of a real-time electrical monitoring system using an ESP32 and a PZEM-004T module. The system measured voltage, current, power, energy, and frequency, highlighting practical embedded systems integration and core electrical engineering concepts. Most importantly, the project provided an opportunity to apply knowledge gained throughout undergraduate coursework in a practical and meaningful way.

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